Uttaran Bhattacharya

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Experience

Research Scientist September 2022 – Present
Adobe Inc. *San Jose, CA, USA*

- Working on generation and editing problems for videos and 3D.

Research Intern May 2022 – August 2022
Google LLC *Mountain View, CA, USA*

- Worked on computational models for facial expression synthesis.

Research Intern May 2021 – August 2021
Adobe Inc. *Bangalore, Karnataka, India*

- Worked on developing video editing solutions.

Research Intern May 2020 – March 2021
Adobe Inc. *San Jose, CA, USA*

- Worked on developing video editing solutions.

Education

University of Maryland 2018 – 2022
Ph.D. Computer Science *Advisor: Dinesh Manocha* *College Park, MD, USA*

Indian Institute of Science 2015 – 2017
M.E. System Science and Automation *Advisor: Venu Madhav Govindu* *Bengaluru, India*

West Bengal University of Technology 2011 – 2015
B.Tech. Computer Science and Engineering *Kolkata, India*

Awards and Nominations

- **IAAI 2025 AI Incidents and Best Practices Best Paper Award.** Akash V. Maharaj, David Arbour, Daniel Lee, Uttaran Bhattacharya, Anup Rao, Austin Zane, Avi Feller, Kun Qian, and Yunyao Li. "Evaluation and Incident Prevention in an Enterprise AI Assistant".
- **ACM MIG 2022 Best Paper Award.** Tanmay Randhavane, Uttaran Bhattacharya, Pooja Kabra, Kyra Kapsaskis, Kurt Gray, Dinesh Manocha, and Aniket Bera. "Learning Gait Emotions Using Affective and Deep Features".
- **UMD Invention of the Year Award 2022 Nominee.** Trisha Mittal, Aniket Bera, Uttaran Bhattacharya, Rohan Chandra, and Dinesh Manocha. "Deepfake Detection Tool".
- **UMD Invention of the Year Award 2021 Nominee.** Trisha Mittal, Aniket Bera, Uttaran Bhattacharya, Rohan Chandra, and Dinesh Manocha. "M3ER: Multiplicative Multimodal Emotion Recognition".
- **ACM Multimedia 2021 Best Paper Award Nominee.** Uttaran Bhattacharya, Elizabeth Childs, Nicholas Rewkowski, and Dinesh Manocha. "Speech2AffectiveGestures: Synthesizing Co-Speech Gestures with Generative Adversarial Affective Expression Learning".
- **IEEE VR 2021 Best Paper Award.** Uttaran Bhattacharya, Nicholas Rewkowski, Abhishek Banerjee, Pooja Guhan, Aniket Bera, and Dinesh Manocha. "Text2Gestures: A Transformer-Based Network for Generating Emotive Body Gestures for Virtual Agents".
- **Adobe Research Fellowship 2021.** Adobe Inc.

- **ACM SAP 2019 Best Poster Award.** Tanmay Randhavane, Uttaran Bhattacharya, Aniket Bera, Kyra Kapsaskis, Kurt Gray, and Dinesh Manocha. "Identifying Emotions from Walking using Affective and Deep Features".
- **Dean's Fellowship 2018.** University of Maryland.
- **Outstanding Student Award 2013.** Institute of Engineering and Management (under the West Bengal University of Technology).

Invited Talks

Current Trends in 3D Pose and Motion Generation

- Purdue University *Host: Aniket Bera* Apr 23, 2024
- ACM@UMD Speaker Series *Host: ACM@UMD Student Chapter* Mar 12, 2024

Affective Human Motion Detection and Synthesis

- University of Southern California *Host: Barath Raghavan* Nov 08, 2022

Bibliography

2026

[1] Shivanshu Shekhar, Uttaran Bhattacharya, Raghavendra Addanki, Mehrab Tanjim, Somdeb Sarkhel, and Tong Zhang. "GT-SVJ: Generative-Transformer-Based Self-Supervised Video Judge For Efficient Video Reward Modeling". IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. *[Upcoming]*

2025

[4] Jiaxin Lu, Chun-Hao Paul Huang, Uttaran Bhattacharya, Qixing Huang, and Yi Zhou. "HUMOTO: A 4D Dataset of Mocap Human Object Interactions". IEEE/CVF International Conference on Computer Vision (ICCV), 2025. *[Poster]*

[3] Ting-Hsuan Liao, Yi Zhou, Yu Shen, Chun-Hao Paul Huang, Saayan Mitra, Jia-Bin Huang, and Uttaran Bhattacharya. "Shape My Moves: Text-Driven Shape-Aware Synthesis of Human Motions". IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. *[Poster]*

[2] Zhengyuan Li, Kai Cheng, Anindita Ghosh, Uttaran Bhattacharya, Liangyan Gui, and Aniket Bera. "SimMotionEdit: Text-Based Human Motion Editing with Motion Similarity Prediction". IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. *[Poster]*

[1] Akash V. Maharaj, David Arbour, Daniel Lee, Uttaran Bhattacharya, Anup Rao, Austin Zane, Avi Feller, Kun Qian, and Yunyao Li. "Evaluation and Incident Prevention in an Enterprise AI Assistant". The Thirty-Seventh Annual Conference on Innovative Applications of Artificial Intelligence (IAAI), 2025. *[Oral][Best Paper]*

2024

[6] Pooja Guhan, Uttaran Bhattacharya, Somdeb Sarkhel, Vahid Azizi, Xiang Chen, Saayan Mitra, Aniket Bera, and Dinesh Manocha. "TAME-RD: Text Assisted Replication of Image Multi-Adjustments for Reverse Designing". The 62nd Annual Meeting of the Association for Computational Linguistics (ACL), 2024. *[Findings]*

[5] Uttaran Bhattacharya, Aniket Bera, and Dinesh Manocha. "Speech2UnifiedExpressions: Synchronous Synthesis of Co-Speech Affective Face and Body Expressions from Affordable Inputs". IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 1st Workshop on Human Motion Generation, 2024. *[Full Paper]*

[4] Supreeth Narasimhaswamy, Uttaran Bhattacharya, Xiang Chen, Ishita Dasgupta, Saayan Mitra, and Minh Hoai. "HanDiffuser: Text-to-Image Generation With Realistic Hand Appearances". IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. *[Poster]*

[3] Akash V. Maharaj, Kun Qian, Uttaran Bhattacharya, Sally Fang, Horia Galatanu, Manas Garg, Rachel Hanessian, Nishant Kapoor, Ken Russell, Shivakumar Vaithyanathan, and Yunyao Li. "Evaluation and Continual Improvement for an Enterprise AI Assistant". Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), Workshops on Data Science with Human in the Loop (DaSH), 2024. *[Full Paper]*

[2] Ashmit Khandelwal, Aditya Agrawal, Aanisha Bhattacharyya, Yaman K Singla, Somesh Singh, Uttaran Bhattacharya, Ishita Dasgupta, Stefano Petrangeli, Rajiv Ratn Shah, Changyou Chen, and Balaji Krishnamurthy. "Large Content And Behavior Models To Understand, Simulate, And Optimize Content And Behavior". International Conference on Learning Representations (ICLR), 2024. *[Spotlight]*

[1] Aneesh Bhattacharya, Manas Paranjape, Uttaran Bhattacharya, and Aniket Bera. "DanceAnyWay: Synthesizing Beat-Guided 3D Dances with Randomized Temporal Contrastive Learning". The 38th Annual AAAI Conference on Artificial Intelligence (AAAI), 2024. *[Poster]*

2023

[1] Supreeth Narasimhaswamy, Uttaran Bhattacharya, Xiang Chen, Ishita Dasgupta, and Saayan Mitra. "Text-to-Hand-Image Generation Using Pose- and Mesh-Guided Diffusion". IEEE/CVF International Conference on Computer Vision (ICCV), 7th International Workshop on Observing and Understanding Hands in Action, 2023. *[Extended Abstract]*

2022

[3] Tanmay Randhavane, Uttaran Bhattacharya, Pooja Kabra, Kyra Kapsaskis, Kurt Gray, Dinesh Manocha, and Aniket Bera. "Learning Gait Emotions Using Affective and Deep Features". ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG), 2022. *[Oral][Best Paper]*

[2] Uttaran Bhattacharya, Gang Wu, Stefano Petrangeli, Viswanathan Swaminathan, and Dinesh Manocha. "Show Me What I Like: Detecting User-Specific Video Highlights Using Content-Based Multi-Head Attention". ACM International Conference on Multimedia (ACMMM), 2022. *[Poster]*

[1] Abhishek Banerjee, Uttaran Bhattacharya, and Aniket Bera. "Learning Unseen Emotions from Gestures via Semantically-Conditioned Zero-Shot Perception with Adversarial Autoencoders". The 36th Annual AAAI Conference on Artificial Intelligence (AAAI), 2022. *[Oral]*

2021

[3] Uttaran Bhattacharya, Gang Wu, Stefano Petrangeli, Viswanathan Swaminathan, and Dinesh Manocha. "HighlightMe: Detecting Highlights from Human-Centric Videos". IEEE/CVF International Conference on Computer Vision (ICCV), 2021. *[Poster]*

[2] Uttaran Bhattacharya, Elizabeth Childs, Nicholas Rewkowski, and Dinesh Manocha. "Speech2AffectiveGestures: Synthesizing Co-Speech Gestures with Generative Adversarial Affective Expression Learning". ACM International Conference on Multimedia (ACMMM), 2021. *[Oral][Best Paper Nominee]*

[1] Uttaran Bhattacharya, Nicholas Rewkowski, Abhishek Banerjee, Pooja Guhan, Aniket Bera, and Dinesh Manocha. "Text2Gestures: A Transformer-Based Network for Generating Emotive Body Gestures for Virtual Agents". IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR), 2021. *[Oral][Best Paper]*

2020

[10] Uttaran Bhattacharya, Nicholas Rewkowski, Pooja Guhan, Niall L. Williams, Trisha Mittal, Aniket Bera, and Dinesh Manocha. "Generating Emotive Gaits for Virtual Agents Using Affect-Based Autoregression". International Symposium on Mixed and Augmented Reality (ISMAR), 2020. *[Oral]*

[9] Trisha Mittal, Uttaran Bhattacharya, Rohan Chandra, Aniket Bera, and Dinesh Manocha. "Emotions Don't Lie: A Deepfake Detection Method using Audio-Visual Affective Cues". ACM International Conference on Multimedia (ACMMM), 2020. *[Poster]*

[8] Uttaran Bhattacharya, Christian Roncal, Trisha Mittal, Rohan Chandra, Kyra Kapsaskis, Kurt Gray, Aniket Bera, and Dinesh Manocha. "Take an Emotion Walk: Perceiving Emotions from Gaits Using Hierarchical Attention Pooling and Affective Mapping". European Conference on Computer Vision (ECCV), 2020. *[Poster]*

[7] Rohan Chandra, Uttaran Bhattacharya, Trisha Mittal, Aniket Bera, and Dinesh Manocha. "CMetric: A Driving Behavior Measure Using Centrality Functions". IEEE/RSJ International Conference on Intelligence Robots and Systems (IROS), 2020. *[Oral]*

[6] Rohan Chandra, Tianrui Guan, Srujan Panuganti, Trisha Mittal, Uttaran Bhattacharya, Aniket Bera, and Dinesh Manocha. "Forecasting Trajectory and Behavior of Road-Agents Using Spectral Clustering in Graph-LSTMs". Robotics and Automation Letters (RA-L), 2020, IEEE/RSJ International Conference on Intelligence Robots and Systems (IROS), 2020. *[Oral]*

[5] Trisha Mittal, Pooja Guhan, Uttaran Bhattacharya, Rohan Chandra, Aniket Bera, and Dinesh Manocha. "EmotiCon: Context-Aware Multimodal Emotion Recognition using Frege's Principle". IEEE/CVF Computer Vision and Pattern Recognition (CVPR), 2020. *[Poster]*

[4] Rohan Chandra, Uttaran Bhattacharya, Trisha Mittal, Aniket Bera, and Dinesh Manocha. "GraphRQI: Classifying Driver Behaviors Using Graph Spectrums". International Conference on Robotics and Automation (ICRA), 2020. *[Poster]*

[3] Rohan Chandra, Uttaran Bhattacharya, Tanmay Randhavane, Aniket Bera, and Dinesh Manocha. "RoadTrack: Realtime Tracking of Road Agents in Dense and Heterogeneous Environments". International Conference on Robotics and Automation (ICRA), 2020. *[Poster]*

[2] Uttaran Bhattacharya, Trisha Mittal, Rohan Chandra, Tanmay Randhavane, Aniket Bera, and Dinesh Manocha. "STEP: Spatial Temporal Graph Convolutional Networks for Emotion Perception from Gaits". The 34th Annual AAAI Conference on Artificial Intelligence (AAAI), 2020. *[Spotlight]*

[1] Trisha Mittal, Uttaran Bhattacharya, Rohan Chandra, Aniket Bera, and Dinesh Manocha. "M3ER: Multiplicative Multimodal Emotion Recognition Using Facial, Textual, and Speech Cues". The 34th Annual AAAI Conference on Artificial Intelligence (AAAI), 2020. *[Oral]*

2019

[4] Uttaran Bhattacharya, and Venu Madhav Govindu. "Efficient and Robust Registration on The 3D Special Euclidean Group". IEEE/CVF International Conference on Computer Vision (ICCV), 2019. *[Poster]*

[3] Rohan Chandra, Uttaran Bhattacharya, Christian Roncal, Aniket Bera, and Dinesh Manocha. "RobustTP: End-to-End Trajectory Prediction for Heterogeneous Road-Agents in Dense Traffic with Noisy Sensor Inputs". ACM Computer Science in Cars Symposium (CSCS), 2019. *[Oral]*

[2] Rohan Chandra, Uttaran Bhattacharya, Aniket Bera, and Dinesh Manocha. "DensePeds: Pedestrian Tracking in Dense Crowds Using Front-RVO and Sparse Features". IEEE/RSJ International Conference on Intelligence Robots and Systems (IROS), 2019. *[Oral]*

[1] Rohan Chandra, Uttaran Bhattacharya, Aniket Bera, and Dinesh Manocha. "TraPHic: Trajectory Prediction in Dense and Heterogeneous Traffic Using Weighted Interactions". IEEE/CVF Computer Vision and Pattern Recognition (CVPR), 2019. *[Poster]*

2017

[1] Uttaran Bhattacharya, Sumit Veerawal, and Venu Madhav Govindu. "Fast Multiview Registration of 3D Scans using Planar Structures". International Conference on 3D Vision (3DV), 2017. *[Spotlight]*

2014

[1] Uttaran Bhattacharya, and Dipannita Dey. "Comparative Analysis of Scheduling Algorithms in Computational Grid Environment". International Journal of Computer Applications 107.4, 2014.

Preprints

[3] Vijeta Deshpande, Ishita Dasgupta, Uttaran Bhattacharya, Somdeb Sarkhel, Saayan Mitra, and Anna Rumshisky. "A Penalty Goes a Long Way: Measuring Lexical Diversity in Synthetic Texts Under Prompt-Influenced Length Variations". arXiv preprint arxiv:2507.15092 (2025).

[2] Aishwarya Sahoo, Jeevana Kruthi Karnuthala, Tushar Parmanand Budhwani, Pranchal Agarwal, Sankaran Vaidyanathan, Alexa Siu, Franck Dernoncourt, Jennifer Healey, Nedim Lipka, Ryan Rossi, Uttaran Bhattacharya, and Branislav Kveton. "Quantitative LLM Judges". arXiv preprint arxiv:2506.02945 (2025).

[1] Yizhou Wang, Ruiyi Zhang, Haoliang Wang, Uttaran Bhattacharya, Yun Fu, and Gang Wu. "VaQuitA: Enhancing Alignment in LLM-Assisted Video Understanding". arXiv preprint arxiv:2309.00359 (2023).

Patents

[5] Trisha Mittal, Aniket Bera, Uttaran Bhattacharya, Rohan Chandra, Pooja Guhan, Dinesh Manocha. "Human Emotion Recognition in Images or Video". Patent US11861940B2. 2024.

[4] Trisha Mittal, Aniket Bera, Uttaran Bhattacharya, Rohan Chandra, Dinesh Manocha. "System and Method for Multimodal Emotion Recognition". Patent US11830291B2. 2023.

[3] Uttaran Bhattacharya, Abhishek Banerjee, Aniket Bera, Pooja Guhan, Dinesh Manocha, Nicholas Rewkowski. "Neural Networks for Generating Emotive Gestures for Virtual Agents". Patent US20230135769A1. 2023.

[2] Gang Wu, Uttaran Bhattacharya, Stefano Petrangeli, Viswanathan Swaminathan."Highlight Video Generated With Adaptable Multimodal Customization". Patent US11574477B2. 2023.

[1] Trisha Mittal, Aniket Bera, Uttaran Bhattacharya, Rohan Chandra, Dinesh Manocha. "System and Method for Detecting Fabricated Videos". Patent US20220138472A1. 2022.

Public Datasets

Multimodal Content

- HUMOTO — 4D human-object interactions + text — https://adobe-research.github.io/humoto/
- IMAD — Images + edit descriptions + edited images — https://tinyurl.com/imagemultiadjust

Services

Area Chair

- AAAI Conference on Artificial Intelligence (AAAI) *2026*

Program Chair

- IEEE/CVF International Conference on Computer Vision (ICCV) Workshop on Short-Form Video Understanding *2025*

Senior Program Committee Member

- AAAI Conference on Artificial Intelligence (AAAI) *2023 – 25*